Analysis Pipeline — Quick Guide

A MATLAB app for analyzing single-molecule binding/FRET data: detects binding events from a state-assignment file, then offers twelve modules covering kinetics, FRET distributions, and conformational statistics.

Running the app

Requires MATLAB with the **Curve Fitting** and **Statistics and Machine Learning** Toolboxes. The folder with `AnalysisPipelineApp.m` and the `+modules` / `+pipeline` folders must be on the MATLAB path — either `cd` there for the session, or (recommended) add it permanently via **Home → Set Path...**. If this app lives inside your lab's shared `MatlabScripts` directory, make sure `MatlabScripts` has been added via **Set Path...** so the app is callable from any Current Folder. Then in the Command Window run: `AnalysisPipelineApp` — this opens the GUI (no arguments needed). Use `app = AnalysisPipelineApp();` instead if you want a handle to the running app for scripting.

Getting started

1. **Load data.** Click **Load State-Assignment File...** and pick a `*_assigned_corrected.mat` file (must contain `states`, `ttotal`, `channelNames`, `nChPerMol`; older files with only `ttotal` and `states` open as 2-color data). The app detects binding events per channel automatically.
2. **(Optional) Set Min event frames** to drop events shorter than a chosen number of frames — useful for filtering out one-frame noise spikes.
3. **Click a module button** in the left sidebar (**Analysis** or **Plotting**) to open its tab.
4. **Adjust the controls** at the top of the tab (Channel, Events, etc.) — most modules recompute live as you change them.
5. **Fit, export a plot, and/or export a CSV** as needed (see below).

Fit vs. Save Fit vs. Export CSV

Modules with curve fitting (exponential or Gaussian-mixture) share one pattern: **Fit** is a *trial* — it draws the curve and shows the fitted parameters but logs nothing. **Save Fit** re-runs the same fit and *also* runs a bootstrap confidence interval, then appends one row to that module's own results table. **Export CSV...** saves that table. Every module's table is independent: no module's fits ever mix into another module's export.

Confidence intervals

Most parameters carry two possible CIs: a **Fit CI** (from the curve fit's own covariance estimate) and a **Bootstrap CI** (from resampling molecules with replacement many times and refitting — see the User Guide for why resampling is done by *molecule*, not by event). Both are 95% intervals, reported in CSVs as a single `[lower, upper]` column.

Modules

Module	Category	What it does
Binding Lifetime Kinetics	Analysis	Single/double-exponential fit to the eCDF of event lifetimes
Association Kinetics	Analysis	Same fit, applied to first-association / reassociation wait times
FRET Efficiency Distribution	Analysis	1–4 component Gaussian-mixture fit to a pooled FRET histogram
FRET State Lifetime	Analysis	Same exponential-fit machinery as Binding Lifetime, split by Low/High FRET state
Association/Dissociation Conformations	Analysis	Does FRET state predict dissociation risk or onset composition? (mixed-effects logistic regression + bootstrap)
Event Sorting by Mean FRET	Analysis	Splits events into Low/High groups by mean FRET; compares group lifetimes (points + bootstrap CI + rank-sum test)
Event Sorting by Lifetime	Analysis	Splits events into Short/Long groups by lifetime; compares group FRET distributions (Gaussian-mixture fits)
Donor-Acceptor Correlation	Analysis	Within-event donor/acceptor intensity Pearson r as a FRET sanity check (histogram, points + bootstrap CI, pooled D-vs-A scatter)
Post-Sync FRET Heatmap	Plotting	Time × FRET-efficiency density heatmap, traces aligned to binding onset
FRET Evolution	Plotting	Individual per-event FRET trajectories right after binding onset
Combined Dataset Overlay	Plotting	Overlays previously-exported `.fig` plots from different sessions/samples into one figure
Summary Figure	Plotting	Auto-detects every module's previously-exported `.png` plots in a chosen folder and composites them into one multi-panel page for a fast overview

Exporting plots

Every analysis module has an **Export Plot...** button producing a publication-style `.eps` + `.png` + `.fig` set (square or landscape panel, transparent background, consistent fonts/line weights) via a shared house style — `.fig` files exported this way can later be combined in **Combined Dataset Overlay**, and the `.png` files can be pulled together automatically by **Summary Figure**.

For details on the statistics, exact CSV columns, and how to interpret each module's output, see the full User Guide.